

SANYAM KAUL

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PROFILE SUMMARY

Software Engineer with an M.Tech in Computer Science from IIT Hyderabad and 3+ years of experience in building scalable production systems and applied AI. Skilled in architecting RAG-based LLM systems, optimizing distributed data pipelines, and developing high-performance financial platforms. Proven track record of reducing cloud costs by 15% and optimizing system latency by 6x.

EDUCATION

M.Tech, Computer Science
IIT Hyderabad 2025
CGPA: 8.90

B.Tech, CSE (Sec & Networks)
VIT-AP University 2021
CGPA: 8.90

TECHNICAL SKILLS

- **Languages:** Java, Python, SQL, C++, Golang, Rust, Javascript, Shell/Bash.
- **AI/ML:** LLMs (RAG, Fine-tuning, PEFT, LoRA/QLoRA), Transformers, LangChain, LangGraph, VAE/Anomaly Detection, Reinforcement Learning.
- **Backend:** Spring Boot, FastAPI, REST, gRPC, GraphQL, Node.js, Microservices.
- **Data & Infra:** Cloud (AWS, Azure), EKS, Lambda, SQS, SNS, S3, Docker, Kubernetes.
- **Databases:** PostgreSQL, MongoDB, ChromaDB, Elasticsearch.
- **Tools:** Git, Jenkins, GitHub Actions, CI/CD, Grafana, Datadog, Prometheus.

PROJECTS

Fraud Detection (VAE) — *Anomaly detection*
[GitHub](#)

- Built VAE model for credit card fraud detection (91%+ accuracy).
- **Tech Stack:** Python, PyTorch, VAE, Scikit-Learn.

Car Insurance OCR — *Structured Data Extraction POC*

- Developed Python system to extract structured data from insurance PDFs.
- **Tech Stack:** OpenAI, Tesseract OCR, FastAPI.

LooksMaxing AI — *AI personalization* [Link](#)

- Built recommendation system using Gemini for virtual try-on.
- **Tech Stack:** Python, FastAPI, Gemini API.

Synthetic Data (VAE) — *VAE Latent Space*

- Generated datasets using VAE latent space exploration.
- **Tech Stack:** Python, VAE, NumPy, Pandas.

EXPERIENCE

Netradyne Bengaluru
Data Engineer (ML Systems) Jul 2025 - Present

- Designed and implemented AWS Monitoring System (Python, MongoDB, FastAPI) for machine metrics and costs (**10–15% cost reduction**).
- Optimized Video Reprocessing System (EKS, SQS, SNS, ASGs), reducing latency from **2 hours to 20 mins** for 50K+ video dataset creation.
- Built scalable infrastructure for large-scale video feature engineering.
- **Tech Stack:** Python, FastAPI, MongoDB, AWS (EKS, S3, SQS, SNS, CloudWatch), Dash.

WinesecretHK Remote
Founding Member (AI Architect) Apr 2024 - Jun 2025

- Built **RAG AI Pairing System** using Python, AWS RDS, and ChromaDB for flavour-profile matching (not just semantic search).
- Developed **HyDE-inspired RAG algorithm** for wine pairing (Patent submitted).
- Implemented multi-modal agentic workflows using LangChain and LangGraph for complex user interactions.
- Integrated Stripe payments and POS webhooks for ordering.
- **Tech Stack:** Python, FastAPI, AWS RDS, ChromaDB, LangChain, LangGraph, Stripe.

BlackRock Gurugram
Tech Analyst (SDE-1) Jul 2021 - Jun 2023

- Developed full-stack Financial Information Reporting Tool (Spring Boot, Angular), reducing setup from **1.5 months to 2-3 days**.
- Standardized REST/GraphQL workflows across **50+ teams**.
- Built **Azure DevOps** analytics dashboard and improved backend test coverage to **85%+**.
- **Tech Stack:** Java (Spring Boot), Angular, REST, GraphQL, Golang, Azure DevOps.

RESEARCH & PUBLICATIONS

HyperLedger Fabric Optimization
M.Tech Thesis — IIT Hyderabad [arXiv](#)

- Optimized architecture to reduce failure rates. Presented at Indosys 2025.

HACKATHONS & ACHIEVEMENTS

- **1st Place (IITH FoML):** Ensemble model (Random Forest/GBDT).
- **15th/100+ (IITH DL):** Advanced DL using ResNet, ViT.
- **Deep Learning Architectures:** Implemented CNN, RNN, Transformers from scratch (NumPy).